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$$\int \frac{1}{x\sqrt{1-x^2}}dx \overset{x=\sin t}{=} \int \frac{1}{\sin t\sqrt{1-\sin^2 t}}\cos t dt = \int \frac{1}{\sin t\sqrt{\cos^2 t}}\cos t dt$$
$$= \int \frac{1}{\sin t}dt \overset{u=\tan\left(\frac{t}{2}\right)}{=} \int \frac{1}{\frac{u}{1+u^2}}\frac{2}{1+u^2}du = \int \frac{1}{u}du = \ln|u| + c$$
$$\overset{u=\tan\left(\frac{t}{2}\right)}{=} \ln\left|\tan\left(\frac{t}{2}\right)\right| + c \overset{t=\text{Bgsin}\, x}{=} \ln\left|\tan\left(\frac{\text{Bgsin}\, x}{2}\right)\right| + c$$

References